

INFORMATION FOR FIRST AND SECOND RESPONDERS

EMERGENCY RESPONSE GUIDE



Hyundai Nexo FCEV

H2 and Li-ion Battery



## Overview

Hyundai Motor Group (HMG)'s Emergency Response Guide contains emergency response measures, warnings and precautions for vehicles. This publication aims to provide information necessary for extrication operations and training for first and second responders – but not for car dealerships, end-users, or any other readers.

This Guide can be constantly updated by the HMG.

This Guide only applies to the All-new NEXO, and contains location and description of high-voltage components and information about vehicle extrication.

However, it does not cover every emergency scenario that may occur. If you do not follow the recommended emergency response procedure, it may result in severe injuries or death.

Since this Guide contains characteristics of the vehicle or any other information you need to know in case of emergency, we recommend you read it carefully in advance.

## Important Information

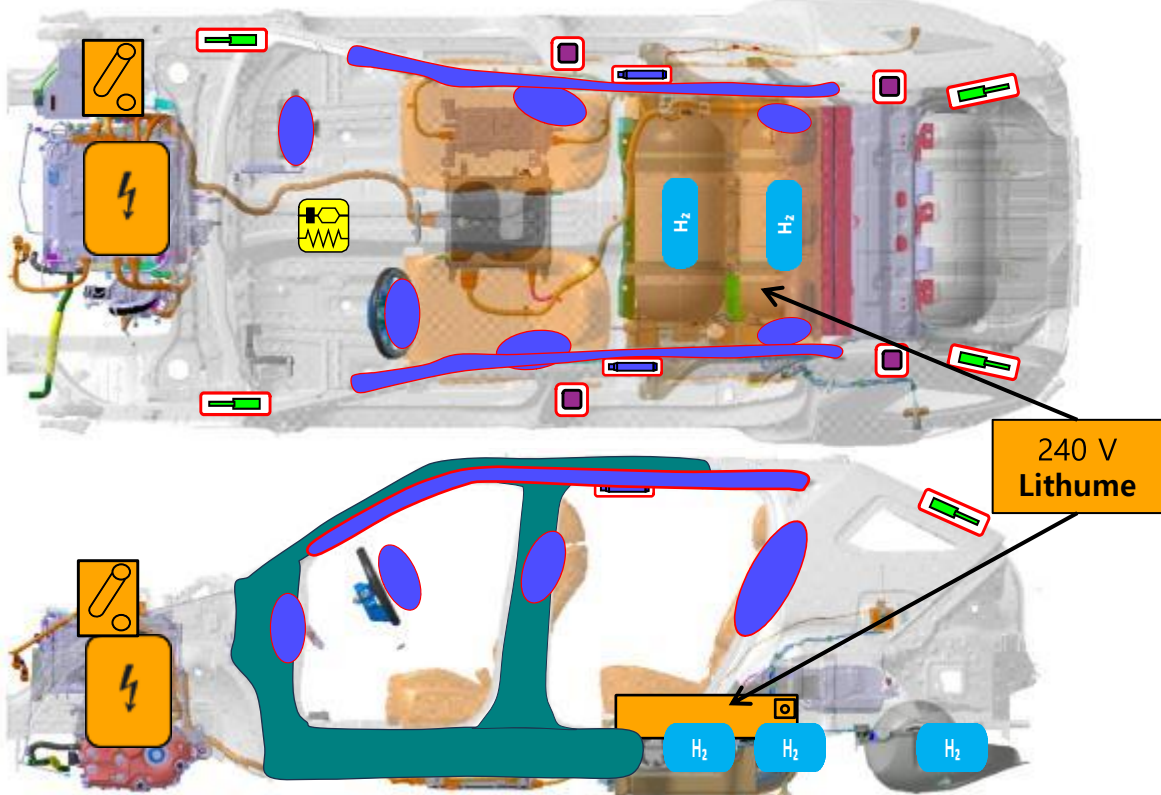


### WARNING

The "WARNING" sign indicates that failure to follow instructions under the Warning section may result in severe damage, bodily injury, or death.

# Table of Contents

|                                                |       |
|------------------------------------------------|-------|
| 0. Rescue sheet(s)                             | 1     |
| 1. Identification / recognition                | 2~9   |
| 2. Immobilization / stabilization / lifting    | 10~11 |
| 3. Disable direct hazards / safety regulations | 12~15 |
| 4. Access to the occupants                     | 16~20 |
| 5. Stored energy/ liquids/ gases/ solids       | 21~25 |
| 6. In case of fire                             | 26~31 |
| 7. In case of submersion                       | 32    |
| 8. Towing / transportation / storage           | 33    |
| 9. Important additional information            | 34~36 |
| 10. Explanation of pictograms used             | 37    |

|                                                                                     |                                |                                                                                     |                              |                                                                                     |                                                                                    |                                                                                     |                                                                                     |                                                                                       |                                   |
|-------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------|
|     |                                | <div>Hyundai Nexo</div> <div>SUV, 5 doors</div> <div>From NOV 2025~</div>           |                              |                                                                                     |  |                                                                                     |  |                                                                                       |                                   |
|    |                                |                                                                                     |                              |                                                                                     |  |                                                                                     |                                                                                     |                                                                                       |                                   |
|  |                                |                                                                                     |                              |                                                                                     |                                                                                    |                                                                                     |                                                                                     |                                                                                       |                                   |
|  | Airbag                         |  | Stored gas inflator          |  | Seat belt pretensioner                                                             |  | SRS control unit                                                                    |  | High voltage component            |
|  | Battery pack, high-voltage     |  | Gas strut / Preloaded spring |  | High strength zone                                                                 |  | High voltage power cable                                                            |  | Gas tank with gas type indication |
|  | Disconnect high voltage device |                                                                                     |                              |                                                                                     |                                                                                    |                                                                                     |                                                                                     |                                                                                       |                                   |
|                                                                                     |                                |                                                                                     |                              |                                                                                     |                                                                                    |                                                                                     |                                                                                     |                                                                                       |                                   |



# 1. Identification / recognition

## Purpose

This Guide is aimed at providing first and second responders with proper measures to deal with the All-new NEXO at an emergency scene. It contains general overview of the vehicle's major system and a range of different circumstances that emergency responders may encounter. The emergency rescue procedure for this vehicle model is partly similar to the procedure for other existing vehicles, and this Guide also provides additional information about how to deal with the high-voltage system.

## Vehicle Description

A hydrogen fuel cell electric vehicle (FCEV) is powered by electricity converted from hydrogen using a fuel cell. Unlike conventional vehicles that use an internal combustion engine fueled by gasoline or other fossil fuels, FCEVs do not require fuel sources since they are powered by the motor that uses electricity generated from an electrochemical reaction between hydrogen and oxygen, and that makes them environment friendly and free from exhaust gas. The regenerative braking system activates when decelerating or driving on a downhill to charge the high-voltage battery, thereby minimizing energy loss and increasing the ready to drive distance.

A FCEV's powertrain is almost the same size and location as an internal combustion engine of conventional vehicles.

1. Powertrain (stack)
2. Motor
3. High-voltage battery
4. Hydrogen tank



# 1. Identification / recognition

## Fuel Cell Stack



FCEVs, unlike internal combustion engine vehicles, use high-voltage electrical energy generated by a fuel cell stack as a power source. To prevent any high voltage-related accident, you should exercise extra caution when handling FCEVs. Please familiarize yourself with the following safety information:

- 1) A metal chassis and an electro-conductive cover are located inside the fuel cell stack to prevent electrocution from the high-voltage components. The fuel cell stack has a high IPXXB rating.
- 2) Stack modules and high-voltage buses, which generate 400 V DC or higher in the fuel cell stack, are designed to maintain reliable insulation resistance with an electro-conductive cover.

When the insulation resistance is lower than the specified value, the fuel cell system notifies it to the user and limits the output current of the fuel cell stack.

## High-voltage Battery System



The high-voltage battery system supplies energy during acceleration and stores energy generated during regenerative braking.

- 1) Located in the cargo room, the high-voltage battery system is protected with a steel case.
- 2) It consists of the total of 128 cells, and each cell is protected by an aluminum case to prevent any electrolyte leakage. If a battery module is damaged, electrolyte may leak out of the battery.
- 3) For safety reasons, overcurrent protection and ceramic coating isolation layers are installed.
- 4) Non-flammable electrolyte is used to prevent explosion and fire in a car accident or other emergency situations.
- 5) High-voltage cables (orange color) are used to connect the battery system to the DC converter.
- 6) A high-voltage controller is installed to control high-voltage cables, and high-voltage fuses and safety plugs are used to interrupt electrical sources for safety.

# 1. Identification / recognition

## General Characteristics of Hydrogen

Hydrogen, a gas having the smallest molecular weight, is odorless, colorless, tasteless, non-corrosive, flammable, and diffusible and needs to be handled with caution. But it is not dangerous in any other way and could be less dangerous than other fuels in some aspects. Hydrogen, being the lightest material, does not mix with air and disperses quickly in the atmosphere. It is relatively safe as it does not easily reach concentrations for explosion.

## Characteristics of Hydrogen



Due to its extremely low density, hydrogen tends to rise rapidly in the atmosphere. So, even if it catches fire, the fume quickly rises and dissipates in the air.

But since hydrogen gas leakage may lead to an accident, it should be prevented.

The following are characteristics of hydrogen gas:

- It has the lowest molecular weight and has the smallest molecular size.
- It has lower density and higher buoyancy than other elements.
- It can cause brittleness in some materials, including metals (However, materials chosen for hydrogen applications are not susceptible to brittleness).
- It is colorless, odorless, and tasteless.
- It burns invisibly and without smoke.
- It has lower ignition energy than any other fuels (less than 10% of ignition energy of other fuels).
- It has a wide flammability range of 4 % and 75 %.

### [Velocity Ratio of Diffusion and Flow in the air ]

| Gas            | CH <sub>4</sub> | C <sub>3</sub> H <sub>8</sub> | H <sub>2</sub> |
|----------------|-----------------|-------------------------------|----------------|
| Diffusion      | 1.0             | 0.63                          | 3.8            |
| Turbulent Flow | 1.0             | 0.6                           | 2.83           |

## Hydrogen Gas Leak Detection System

The hydrogen gas detection sensors detect a hydrogen gas leak. If there is a leak, the hydrogen storage system and the power system will be shut down and the sensors will ring alarm bells when concentrations reach the minimum flammability level of hydrogen. The sensors are installed at the fuel cell stack, fuel supply system, and between the hydrogen storage tanks to monitor and block a hydrogen leak.

# 1. Identification / recognition

The all-new NEXO uses high voltage of approximately 180-400 VDC and high-pressure hydrogen gas. Be sure to follow the safety instructions below. Failure to follow the safety instructions may result in serious injury or death.

## [Safety Precautions for the Hydrogen

### Caution

Hydrogen gas leaks should be prevented due to the possibility of ignition or detonation. The following are properties of hydrogen gas.

- It has the lowest molecular weight, and is the smallest element.
- It has lower density and higher buoyancy than any other elements.
- It can cause brittleness in some materials, including metals (However, materials chosen for hydrogen applications are not susceptible to brittleness).
- It is colorless, odorless, and tasteless.
- It burns invisibly and without smoke.
- It has lower ignition energy than any other fuels (less than 10% of ignition energy of other fuels)
- It has a wide flammability range of 4 % and 75 %.

### Warning



- Ignition sources, including flames, sparks, electrostatic discharge or hot surfaces, may ignite hydrogen gas and should not be near the vehicle.
- Caution labels for hydrogen are attached to the hydrogen storage system components. The hydrogen storage system is composed of three tanks filled with hydrogen gas. Each tank is made of a plastic liner to prevent hydrogen permeation and covered with high-strength carbon fiber, allowing the storage tank to sustain high pressures. The tanks contain flammable gas compressed under high pressure. Improper installation, lack of maintenance or overfilling may result in serious injury or death. Do not attempt to remove this cylinder or any of its fittings from this vehicle. It may contain compressed residual gases, which could cause a fire or explosion.

## [Safety Precautions for the High-Voltage System]

### Warning



- Warning labels for high voltages are attached to the high-voltage components. High-voltage cables and connectors are colored orange. Do not touch any of these high-voltage components, cables, and connectors without proper Personal Protection Equipment (PPE) (e.g. electrical gloves, protection glasses, and electrical safety shoes).

# 1. Identification / recognition

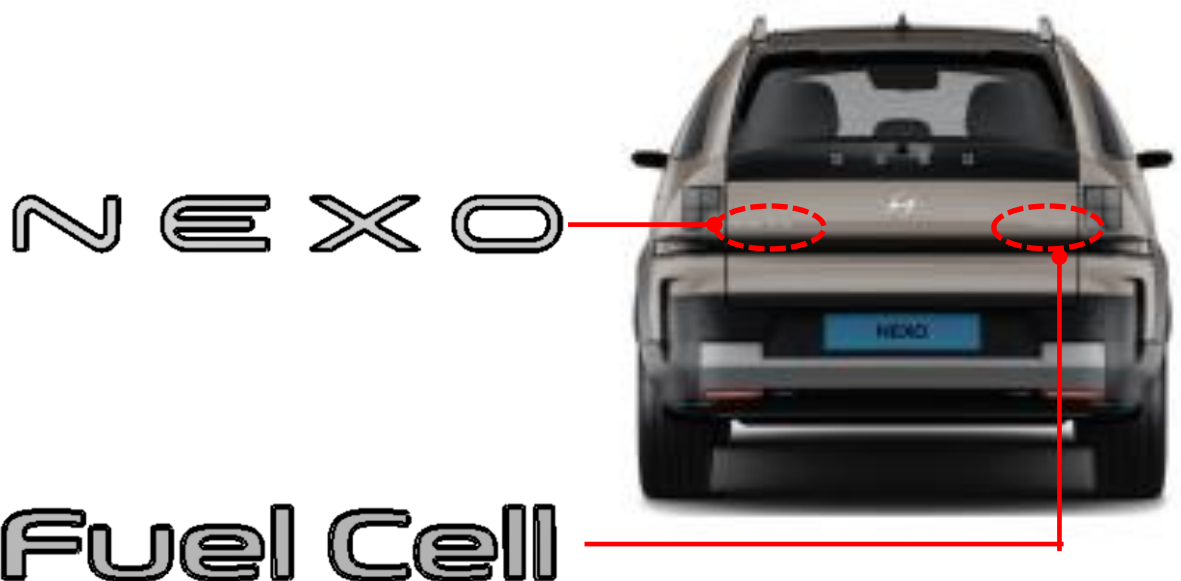
## Vehicle Overview

The all-new NEXO hydrogen electric vehicle is built on a specifically developed chassis. Since it is a hydrogen fuel cell model, the safest method is to assume that any NEXO you handle is equipped with high-voltage and hydrogen systems. A NEXO can be identified by the following identification information.

## Identifying a Vehicle

The all-new NEXO can be easily identified by the "NEXO" and the "Fuel Cell" badges located on the tailgate.

The badges can fall off or become obscured after a crash or damage to the vehicle. Always be sure to check if there is no missing badge.

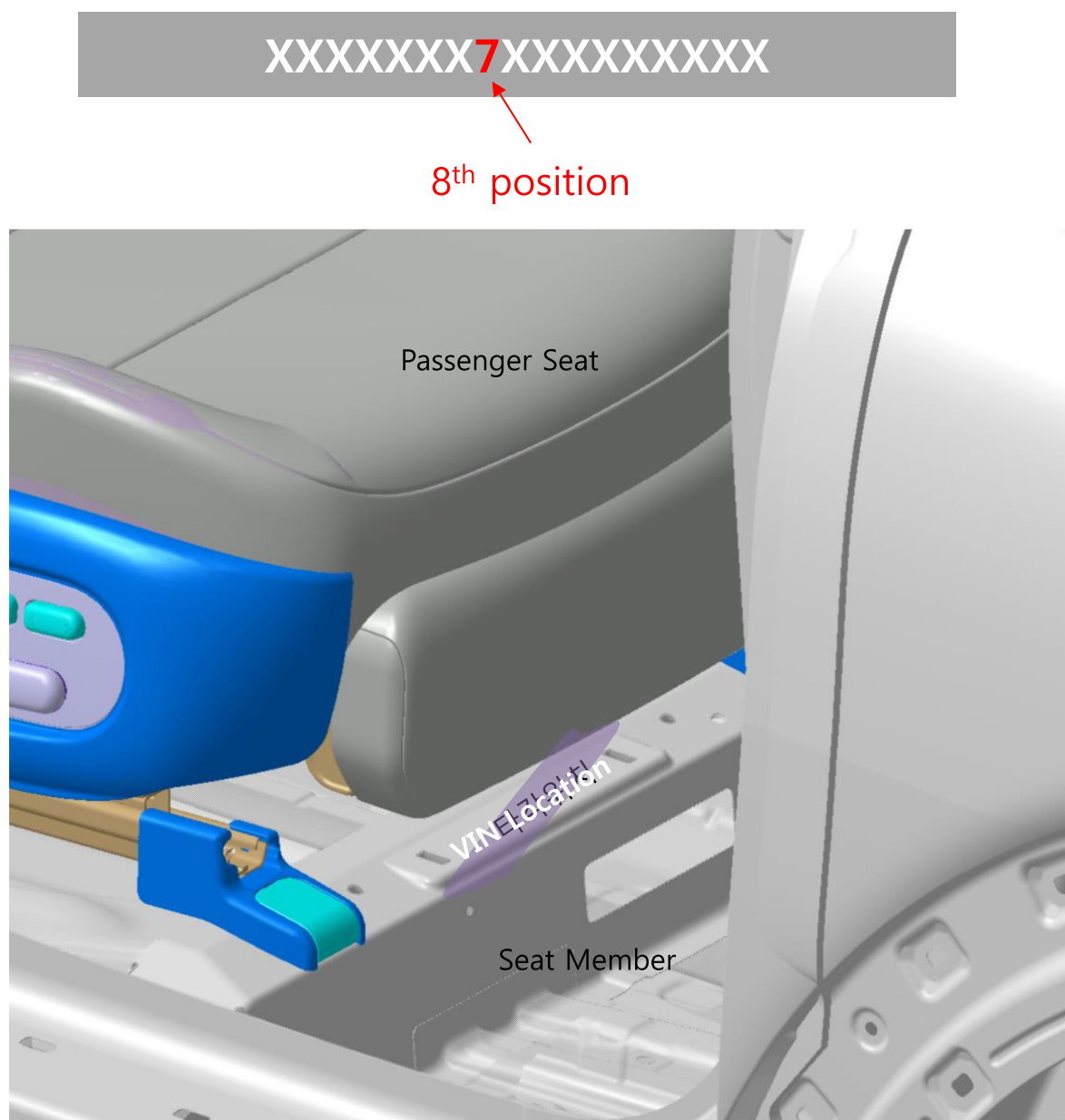


# 1. Identification / recognition

## Vehicle Identification Number (VIN)

The 8th position of the VIN identifies the vehicle as a FCEV.

The VIN is engraved under the passenger seat as in the illustration below. The number "7" in the 8th position of the VIN indicates that it is a FCEV with a battery [LiPB 240 V, 11 Ah] and a motor [3 phase AC 150 kw].



※ The VIN is visible even when the engraved part is covered by a floor mat due to its slit.



# 1. Identification / recognition

## Fuel Cell Stack Room



The All-new NEXO features "HTWO", the trademark of Hyundai Motor Group's hydrogen business brand, clearly marked on its fuel cell stack cover.

Orange-colored high-voltage cables are located in the fuel cell stack room.

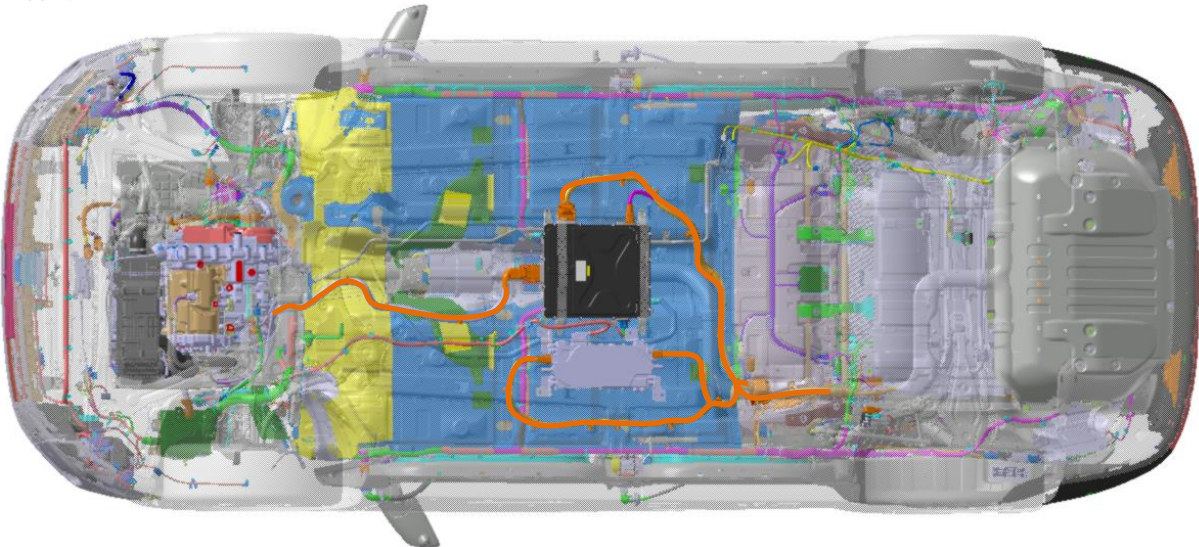


## Vehicle Underbody



The orange-colored cables are also located at the underbody of the vehicle and will become visible if the underbody undercover is removed.

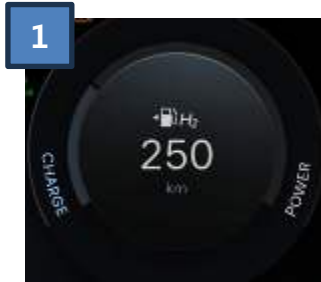
These cables run from the rear part of the vehicle (the part where the high-voltage batteries are located) to the fuel cell stack room.



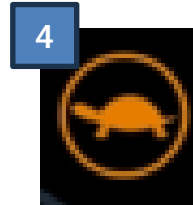
# 1. Identification / recognition

## Cluster Instrument Panel

The instrument cluster panel displays the vehicle's specific features that identify NEXO as a FCEV.



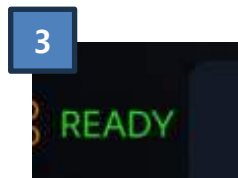
1 : Regenerative braking/ECO/Power gauge



4 : Power Down Warning Light illuminates when the fuel cell stack malfunctions, and the vehicle power supply is limited.



2 : Hydrogen fuel gauge indicates an approximate amount of fuel remaining in the fuel tank.



3 : The "Ready" Light indicates the vehicle is ready for driving.



## 2. Immobilization / stabilization / lifting

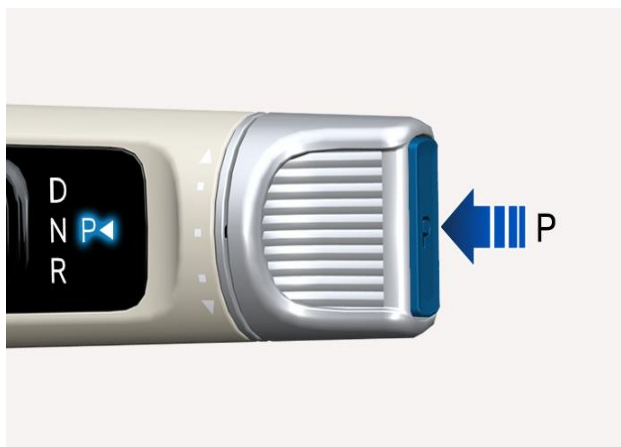
### Immobilization

The next step is to immobilize the vehicle to prevent its accidental movement that may endanger first and second responders and the injured. Since the All-new NEXO is not equipped with a combustion engine, there may be some occasions where it may seem that the vehicle is turned off due to no engine noise.

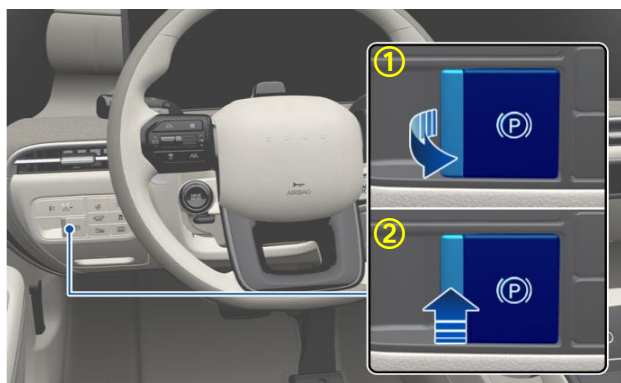
When in the "READY" mode, the vehicle can move silently using only the electric motor.

Emergency responders should approach the vehicle from the sides and stay away from its front and rear since they are potential moving paths of the vehicle.

Follow the steps below to immobilize the vehicle:



Engage the parking brake to stop the vehicle, and press the P button.



Pull up the parking brake switch on the lower left of the driver's side console to activate the parking brake.

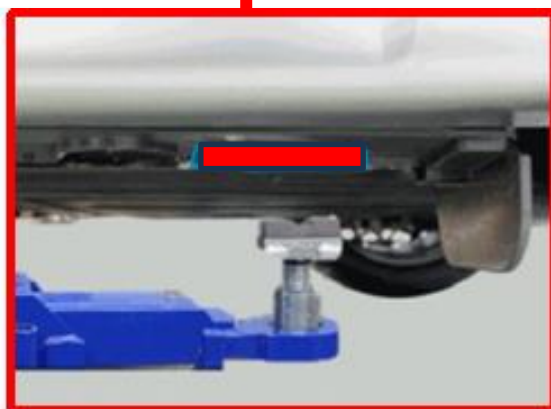
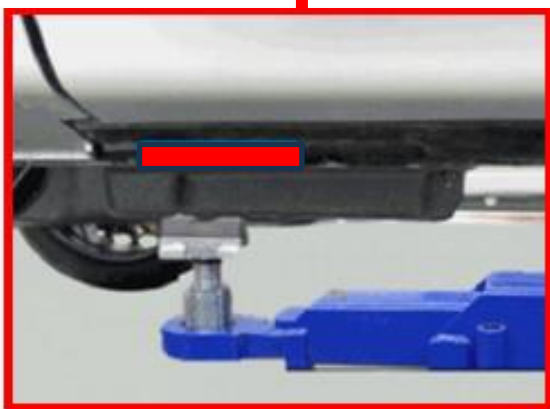
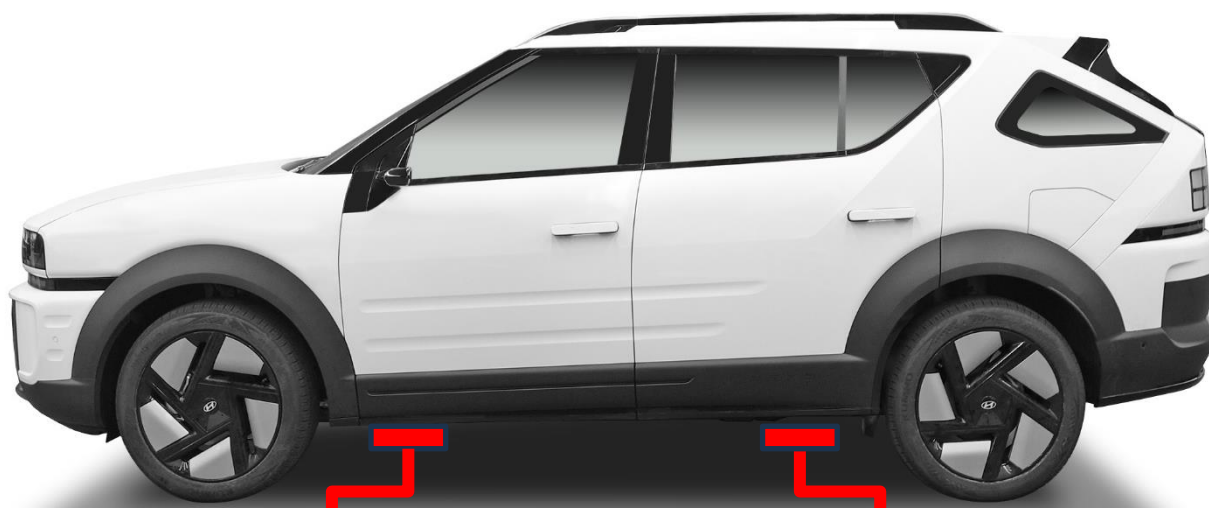


Choke the tires.

## 2. Immobilization / stabilization / lifting

### Lifting

Use the stabilization (lifting) points (  ) at the structural member of the vehicle, as shown in the illustration below. Avoid placing cribs under high-voltage cables and other areas not normally considered acceptable.



※ An actual vehicle may be different from the above illustration.



### 3. Disable direct hazards / safety regulations

#### Deactivating (Disabling) the System



The final stage in the initial response process is to disable the vehicle, its SRS components and the high-voltage electrical system. You can disable the high-voltage electrical system using the following method.

#### I . System Disabling Smart Key System and Power Button

1. Check the status of the Ready to Drive light on the instrument panel.

If the light is on, the vehicle is ON.

- a) If the light is off, the vehicle is OFF.

Do not press the START/STOP button as the vehicle may start moving.

- b) To disable the system, press P (park) on the shift lever and press the EV START/STOP button.



#### Without depressing the brake pedal

| Pressing Power Button | Button Position | Vehicle Condition                                                |
|-----------------------|-----------------|------------------------------------------------------------------|
| -                     | OFF             | OFF                                                              |
| One time              | ACC             | Some electrical devices are operational.                         |
| Two times             | ON              | The warning lights can be checked before the vehicle is started. |
| Three times           | OFF             | OFF                                                              |

#### While depressing the brake pedal and the shift lever in P position

| Pressing Power Button | Button Position | Vehicle Condition |
|-----------------------|-----------------|-------------------|
| -                     | OFF             | OFF               |
| One time              | -               | Ready to drive    |

#### NOTICE

- In an emergency during driving, press and hold the START/STOP button for at least 2 seconds or rapidly press the button more than three times within 3 seconds, and the vehicle will stop and shift to the ACC state. While driving, it is possible to turn on the vehicle again by placing the shift gear in N (Neutral) without depressing the brake. Never turn off the vehicle during normal driving.

### 3. Disable direct hazards / safety regulations

2. Place a smart key at least 2 meters away from the vehicle to prevent it from being accidentally restarted until the 12 V battery is successfully disconnected.



Smart Key

3. To prevent accidental restart of the vehicle, disconnect the (-) connector (A) of the 12 V battery located in the trunk.



#### NOTICE

Before disconnecting the 12 V battery, open the windows, unlock the doors, and open the trunk as required.

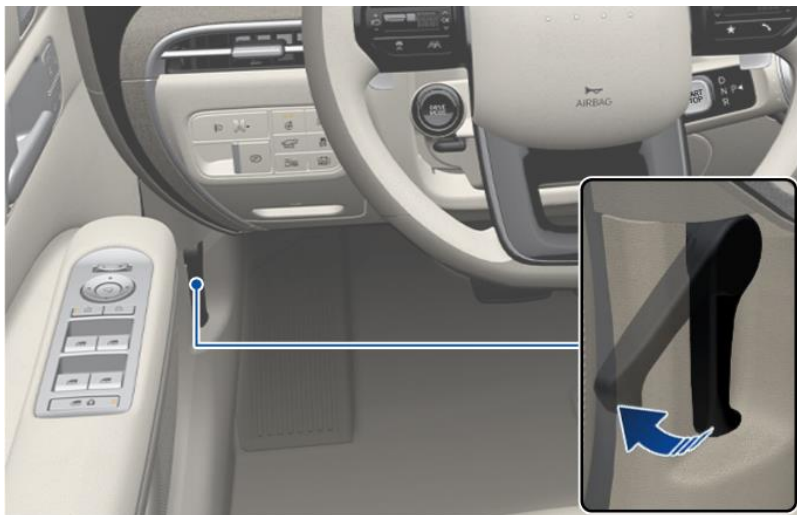
Once the 12 V battery is disconnected, window and door power controls do not operate.



(-) plug terminal located in the trunk

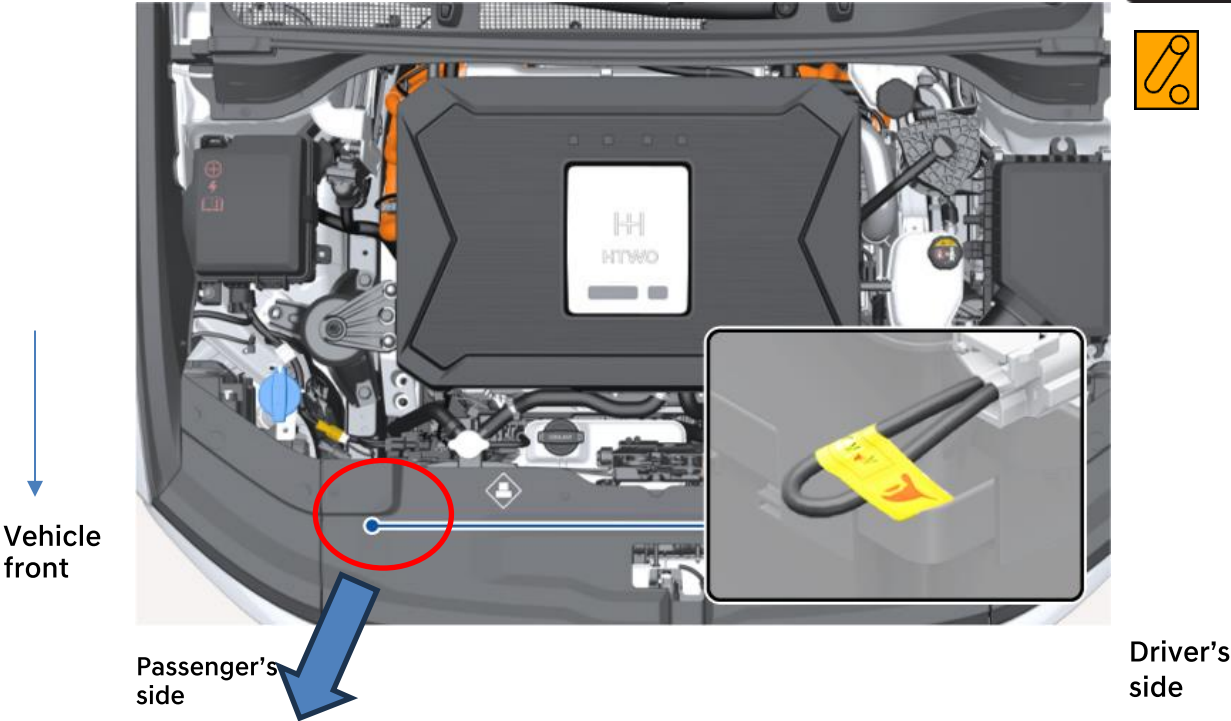
4. Follow the procedure below to disable the high-voltage battery:

- Remove the high-voltage cut-off switch located in the motor room.
- a) Use the hood release lever inside a vehicle to open the hood.

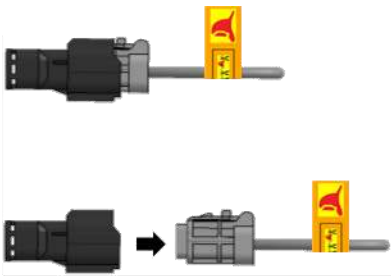


# 3. Disable direct hazards / safety regulations

b) Locate the high-voltage shutoff switch (connector) located in the motor room and at the rear of the vehicle's right-side headlamp.



Locate the emergency shut-off label, and then disconnect the connector or cut the wires.



## WARNING



- If the high-voltage system is not disabled, the 12 V electric system may still be operating even after you disconnect or cut the 12 V battery connector.



### 3. Disable direct hazards / safety regulations

Keep in mind that if the aforementioned method is not possible, there is a likelihood that the SRS system will suddenly activate and the high-voltage system will not shut down.

#### WARNING

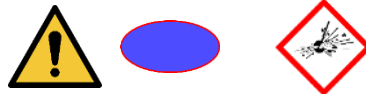
#### Electrocution Risk!



- Make sure to shut down the high-voltage system before starting any emergency operation. Even after removing the safety plug, you have to wait at least 5-10 minutes to allow the high-voltage stored in the capacitor condenser inside the inverter to discharge.
- Exposed cables or wires may be visible inside or outside the vehicle. To prevent severe injury or death from electric shock, never touch the cables or wires before disabling the high-voltage system.

#### WARNING

#### Explosion Risk!



- Sudden activation of the SRS components may result in severe injury or death. Do not cut the SRS components.
- SRS components may remain powered or activated even after the 12 V electrical system shuts down or is disabled. Disconnect the (-) cable of the battery and wait for at least 3 minutes before performing any operations.

## 4. Access to the occupants

### Initial Response

In case of emergency, an initial operation on the All-new NEXO should be conducted in the following procedure.

For operations other than those presented below, we advise that first and second responders follow their standard procedure or guideline when performing such operations.

Keep in mind that if the vehicle is damaged due to an accident, the high-voltage safety system may be compromised and there may be a risk of electrocution. Familiarize yourself with safety precautions, and wear appropriate personal protective equipment (PPE), such as electrical gloves and boots. Remove all metallic jewelry, including watches and rings, before starting operations.

### Identify

When performing an operation on the All-new NEXO in an emergency, the first and second responders should always assume that the vehicle is equipped with the high-voltage system and identify the vehicle model by the VIN as explained in the identification/recognition section of this Guide. Badges on the exterior of the vehicle are visible in most cases, but may be invisible if they fall off or become obscured after a crash or damage to the vehicle. Responders should thoroughly inspect every part of the vehicle by using the vehicle identification mark under the hood and inside the vehicle.

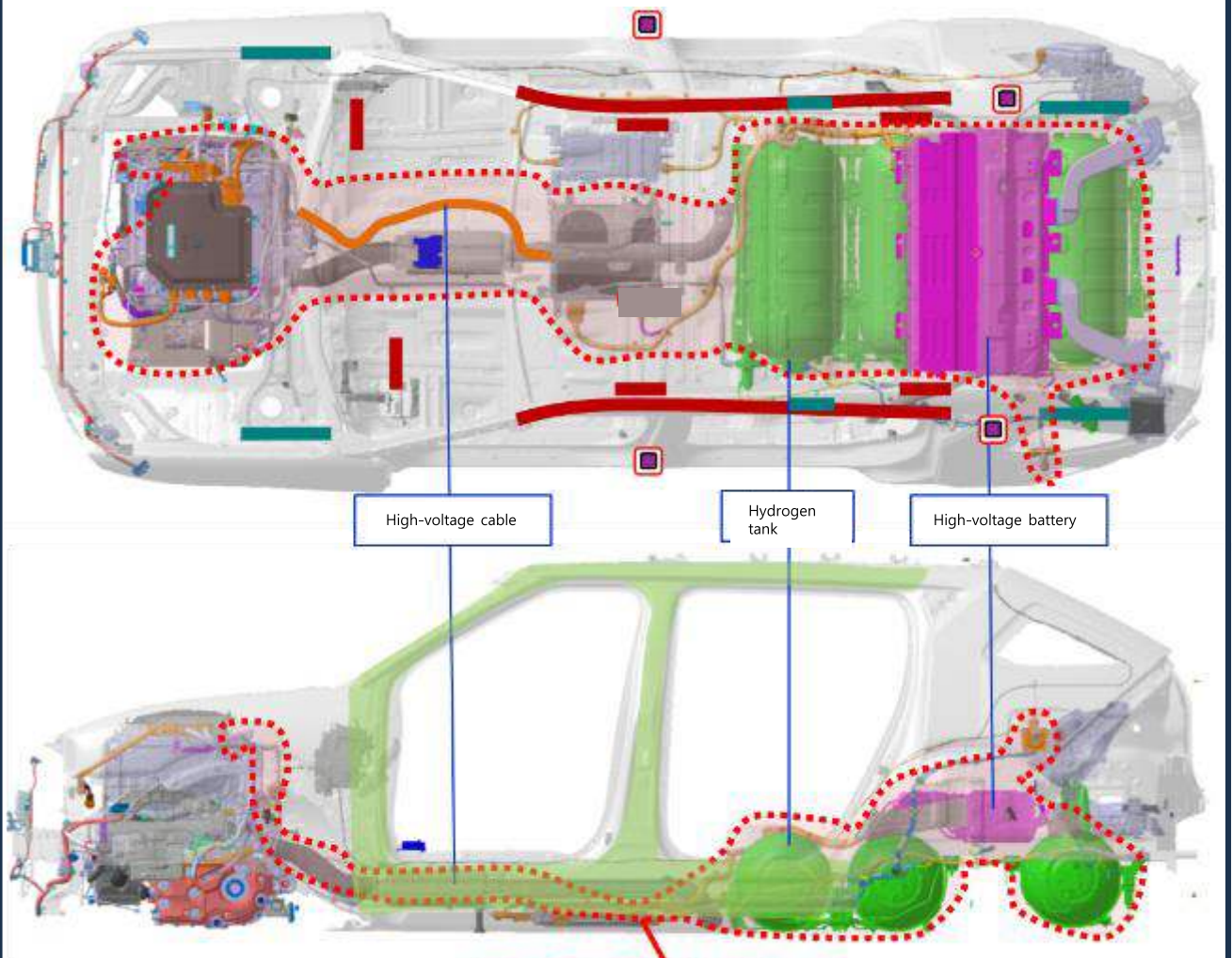
## 4. Access to the occupants

### Rescue Tools and Procedure

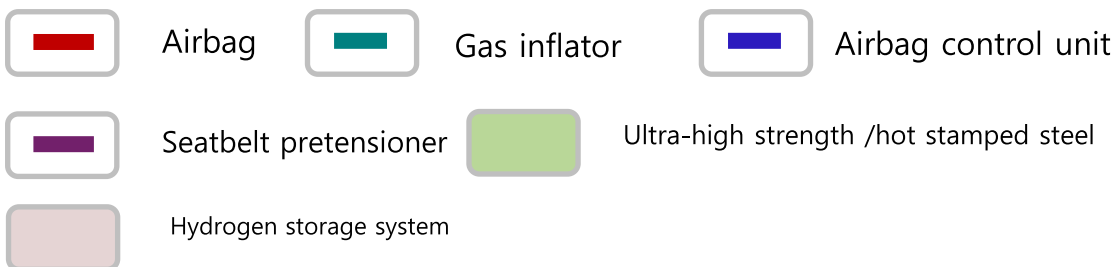
We recommend that first and second responders follow their organization's standard operating procedures when performing an extrication operation on the All-new NEXO.

When cutting the vehicle, emergency responders should take extra caution so that the SRS and the red-colored parts are not cut or broken. The red-colored area indicates the hydrogen storage system and may have an increased likelihood of explosion if damaged.

Ultra-high strength/hot stamped steel is used in the light green-colored areas, and they can be challenging or extremely difficult to cut with general tools.



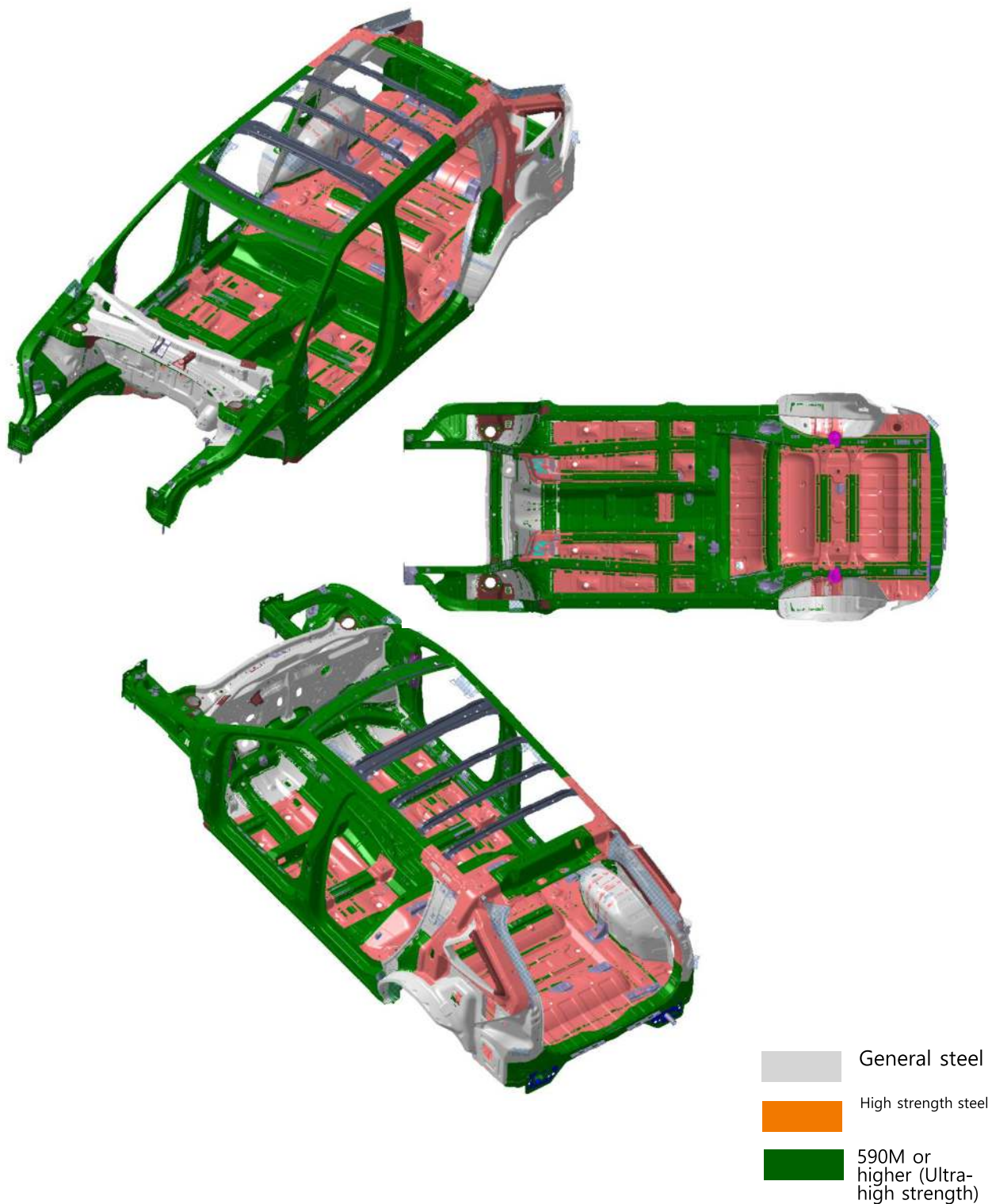
Area that should not be cut (hydrogen storage system)



## 4. Access to the occupants

### Ultra-high Strength Steel

As shown in the illustration below, ultra-high strength steel is used in areas colored green, and high strength steel in areas colored orange. Areas where ultra-high strength steel is used can be challenging or extremely difficult to cut with general tools.



# 4. Access to the occupants

## Extrication

The All-new NEXO extrication operation is almost the same as an extrication procedure performed on other existing vehicles, but there are some notable exceptions that require special attention. Identify, immobilize and disable the vehicle as described in the initial emergency response section before starting an extrication operation.

## Vehicle Stabilization

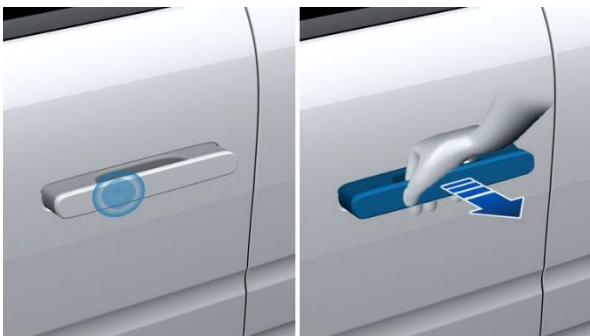
Use standard stabilization (lifting) points as in the picture. Always be sure to connect to the bottom structural member of the vehicle and avoid placing cribs under high-voltage cables and any other areas not normally considered acceptable.



## How to Manually Deploy the Door Handle

(Door handle automatically pops out during airbag deployment, but it may fail to pop out due to a collision.)

1. Push the front part of the handle and the handle pops out.
2. Pull the popped out handle.
3. Pull the handle again to open the door.



## Window Glass Specifications

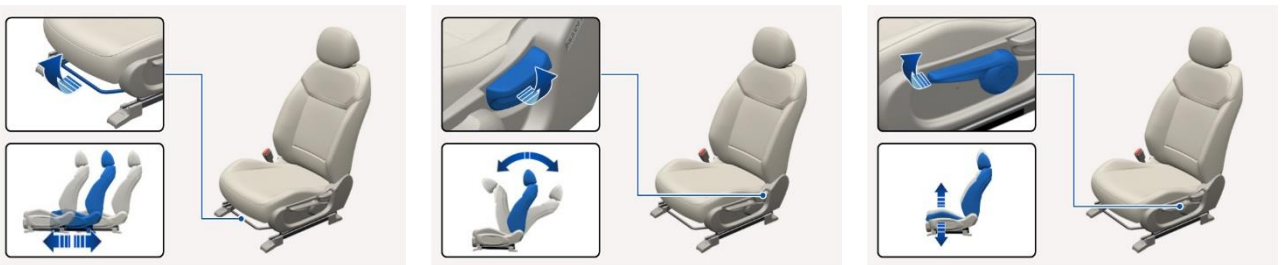


| Glass | Material        |
|-------|-----------------|
| ①     | Laminated glass |
| ②     | Tempered glass  |



## 4. Access to the occupants

### Seat Adjustment



**Manual seat:** Front seat adjustable by a lever

- 1) How to move the seat forward or backward:
  - ① Lift the seat slide adjustment lever upwards.
  - ② Move the seat to the desired position.
  - ③ Check whether the seat is properly fixed.
- 2) How to tilt the seatback backward:
  - ① Lift the seatback lever. ② Move the seatback to the desired position.
  - ③ Check whether the seatback is properly fixed.
- 3) How to adjust the seat cushion height:
  - ① Repeatedly push the lever to lift the seat cushion.
  - ② Repeatedly push the lever to raise the seat cushion.



**Power seat:** Front seat adjustable by a control switch

- 1) How to move the seat forward or backward :
  - ① Push the control switch forward or backward.
  - ② Release the switch once the seat is in your preferred position.
- 2) How to tilt the seatback backward :
  - ① Push the control switch forward or backward.
  - ② Release the switch once the seatback is in your preferred position
- 3) How to adjust the front part of the seat cushion (seat tilt):
  - ① Push the front part of the control switch upward or downward.
  - ② Release the switch once the seat is in your preferred position.
- 4) How to adjust the seat height:
  - ① Push the back part of the control switch upward or downward.
  - ② Release the switch once the seat is in your preferred position.

### Steering Wheel Adjustment

1. Operate the steering wheel lever.
2. You can adjust the steering wheel height.
3. You can adjust the steering wheel reach.

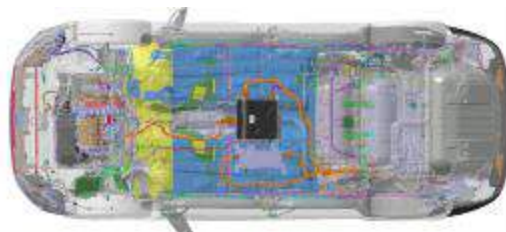




## 5. Stored energy/ liquids/ gases/ solids

### High-voltage Cables ⚡ ⚠

The All-new NEXO's high-voltage cables are in orange color as per the SAE Standards. The cables run under the vehicle's floor from the high-voltage battery to the electric motor and other high-voltage components at the front.



#### WARNING

Electrocution Risk!



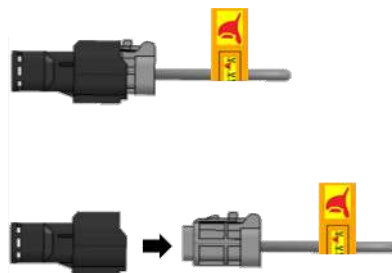
- Never cut the orange-colored high-voltage cables before disabling the high-voltage battery system using the system disabling (deactivation) method as explained earlier.
  - Exposed cables or wires may be visible inside and outside the vehicle. Never touch the cables or wires before disabling the high-voltage system in order to prevent severe injury or death from electrocution.
- ※ Failure to follow the above instructions may result in severe injury or death from electrocution.



Make sure to disable (deactivate) the high-voltage system before performing any operation that involves handling of high-voltage battery or components. Wait for at least 5-10 minutes after deactivating the system to discharge the high-voltage capacitor.



Check the high-voltage shutoff switch (connector) label located in the motor room and at the rear of the vehicle's right-side headlamp, and remove the connector or cut the wire.



## 5. Stored energy/ liquids/ gases/ solids

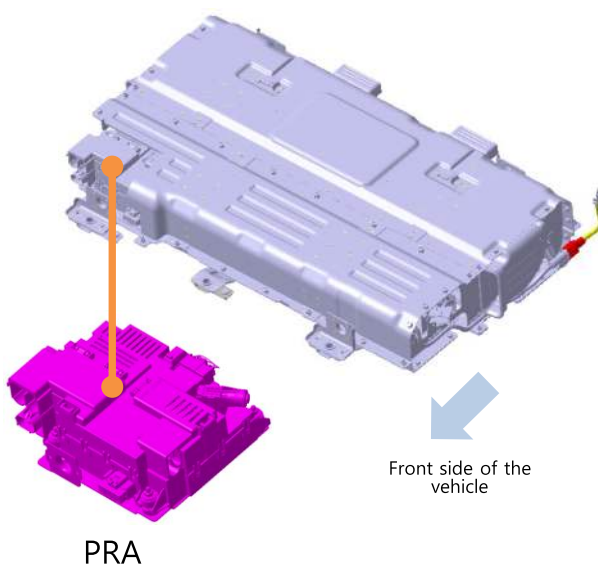
### High-Voltage Electrical Isolation

Unlike the 12 V electrical system that is grounded to the vehicle's chassis, the All New NEXO's high-voltage electrical system is designed to be isolated from the vehicle.

### High-Voltage Electrical Current Control



The high-voltage battery's current is controlled by the Power Relay Assembly (PRA), which consists of the positive (+) and negative (-) main relays, a pre-charge relay, pre-charge resistor, and the battery current sensor. The PRA is located inside the high-voltage battery pack assembly and controls the high-voltage power circuit between the high-voltage battery and the Electric Power Control Unit (EPCU).



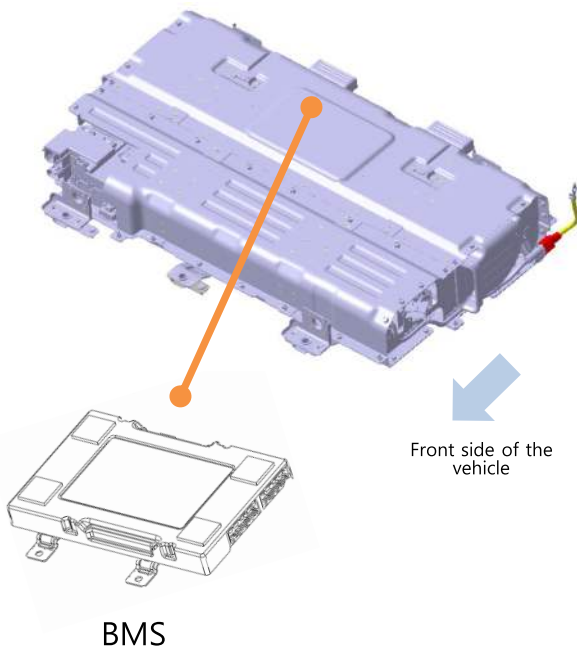
PRA

### High-voltage Safety System



There are multiple safety systems incorporated into the All-new NEXO. The system that protects the high-voltage electrical system is called the Battery Management System (BMS). The BMS is located inside the PRA and measures several parameters to maintain the optimal performance of the high-voltage battery.

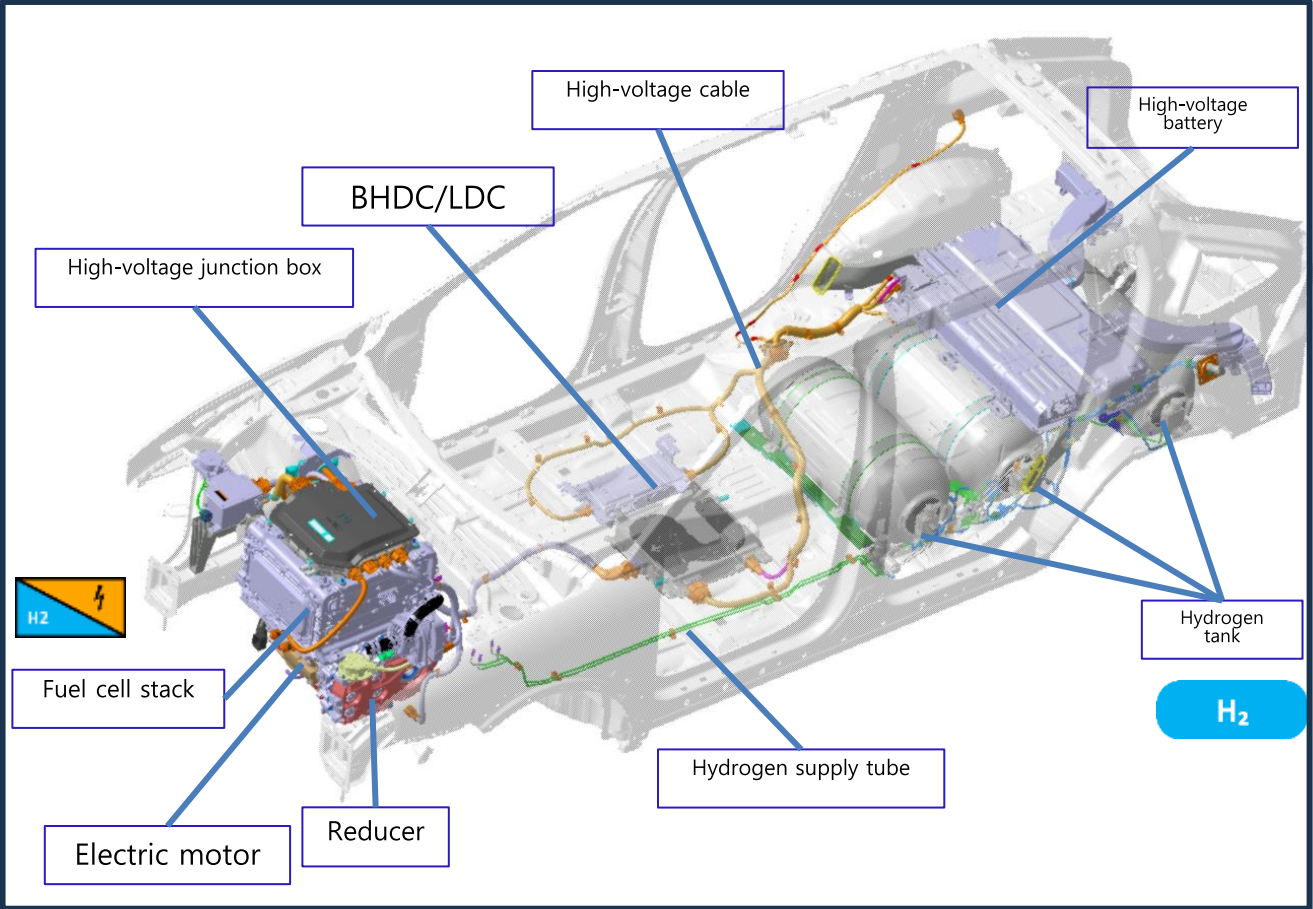
It controls the battery cooling fan to ensure proper battery operation. In addition, if a system fault occurs, the BMS shuts off high voltage power to protect the system.



BMS

# 5. Stored energy/ liquids/ gases/ solids

## Vehicle Component Location



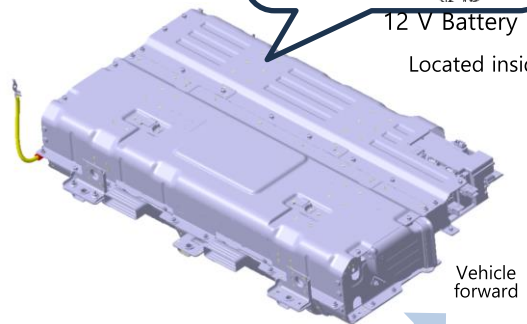
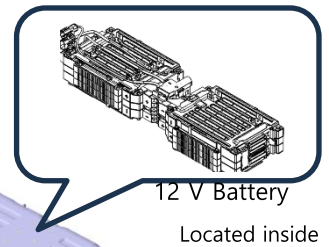
|                           |                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Fuel cell stack           | Generates electricity through the electrochemical reaction of hydrogen and oxygen                                                         |
| Hydrogen tank             | Stores hydrogen fuel, 162.6 L (54.2 L X 3)                                                                                                |
| LDC                       | Low Power DC - DC Converter:<br>Charges 12 V auxiliary battery using electricity from the high-voltage battery                            |
| Inverter                  | DC → AC (Converts fuel cell stack's high-voltage DC to AC to power the electric motor)<br>AC → DC (Charging through regenerative braking) |
| High-voltage junction box | Distributes power generated from a fuel cell to high-voltage components                                                                   |
| BHDC                      | Converts voltages among the high-voltage battery, fuel cell stack, and inverter                                                           |
| Electric motor            | When current flows through the coil, it generates a magnetic field, causing the motor to rotate and produce torque.                       |
| Reducer                   | It increases torque of the electric motor and transmit the increased torque to the vehicle's wheels.                                      |
| High-voltage battery      | It supplies electrical current to the motor and stores electricity generated.                                                             |

## 5. Stored energy/ liquids/ gases/ solids

### Vehicle Components

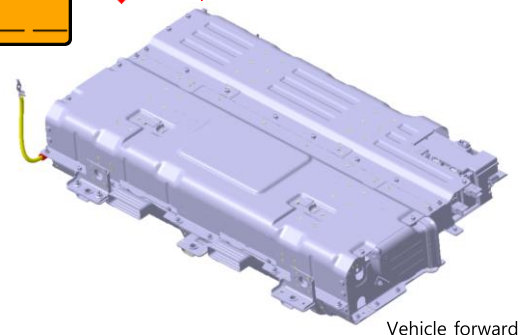
#### 12 V Battery

The 12 V lithium-ion polymer auxiliary battery is located at the front of the high-voltage battery case located inside the trunk and supplies power to electrical devices, such as radio and instrument panel. It also supplies power to a control unit that controls the high-voltage current flow and hydrogen supply.



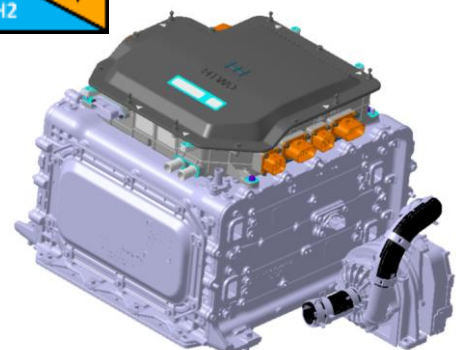
#### High-voltage Battery

Located inside the trunk, the lithium-ion polymer high-voltage battery contains gel electrolyte and consists of 128 3.75 V cells serially connected for a 240 V nominal voltage and an 11 Ah capacity.



#### Fuel Cell Stack

The fuel cell stack consists of a stack that generates electricity through electrochemical reaction between hydrogen and oxygen, a hydrogen supply system to supply the hydrogen fuel, a thermal management system that controls heat, and an oxygen supply system. Electricity generated from the fuel cell system, along with the high-voltage battery, powers an electric motor to drive the vehicle.





## 5. Stored energy/ liquids/ gases/ solids

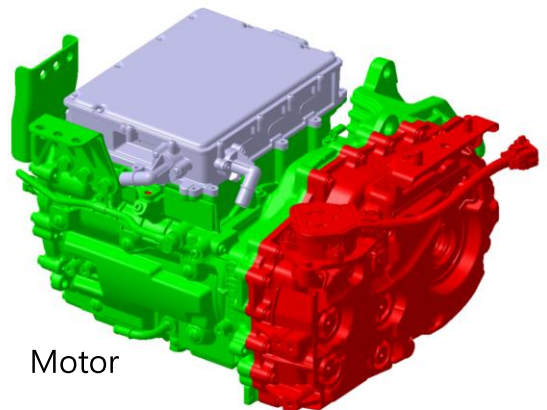
### Motor Control Unit

The inverter converts the direct current (DC) of the high-voltage battery to the alternating current (AC) and supplies it to the motor, and also converts the AC of the regenerative braking to the DC to recharge the high-voltage battery. The inverter is equipped with a Motor Control Unit (MCU) to control the motor torque. When dealing with the high-voltage components, wait 5 to 10 minutes to allow the high-voltage capacitor in the inverter to discharge.



### Electric Motor

Mounted in the fuel cell stack room with the speed reducer, the electric motor is used for vehicle propulsion. During deceleration or braking, it acts as an alternator and charges the high-voltage battery by converting the vehicle's kinetic energy into electrical energy.



Motor

Speed  
Reducer

### Speed Reducer

It increases motor torque and transmits the increased torque to the vehicle wheels.



## 6. In case of fire

### Emergency Venting of Hydrogen Gas



If the temperature near the safety valve located at the rear of the vehicle underbody exceeds 110°C due to a fire or other reasons, the safety valve will open to vent hydrogen gas. Complete venting of hydrogen gas takes approximately 300 seconds and may make a loud noise when the gas is released at high speeds. Stay far away from the vehicle, as the released hydrogen gas may produce a jet flame that ignites.



Hydrogen venting position and direction

### WARNING



- Stay far away from the vehicle for at least 5 minutes during hydrogen gas venting.

## 6. In case of fire



### ● Fuel Cell Electric Vehicle (FCEV) Fire Characteristics and Precautions

Since FCEV fires have characteristics different from internal combustion engine vehicle (ICEV) fires, a special care is required when extinguishing such fires. The high-voltage battery pack in FCEVs makes those fires present a greater risk and there is also a risk of explosion and reignition. Please refer to the following key precautions concerning FCEV fires.

#### • High-voltage Electricity System

Since FCEVs use the high-voltage battery system, make sure to shut down the high-voltage system before approaching a fire scene. Locate the high-voltage battery pack, wires and electrical components and wear appropriate Personal Protective Equipment (PPE), such as electrical gloves and boots, to avoid an electrocution risk when approaching the scene.

#### • Risk of Battery Fire

If an FCEV catches fire, its lithium-ion batteries may explode and there is a risk that the fire may spread. When lithium-ion batteries are exposed to the temperature of 300 F /148 °C or higher, electrolytes may decompose, leading to gas emissions. Fire may break out again if the gas is ignited by a flame, spark, or other ignition sources.

#### • Use a Large Volume of Water

When suppressing FCEV fires, it is important to lower the temperature of the high-voltage battery cell by using a large volume of water. The amount of water required for putting out such fires may be much more than that required for suppressing ICEV fires. Since fire-extinguishing agents other than water (e.g. CO<sub>2</sub> fire extinguisher, powder extinguisher, etc.)

have no effect in suppressing EV battery fires, do not use such fire-extinguishing agents.

#### • Set a Danger Zone

Prohibit the access of persons other than emergency responders to the fire scene and set an area having a possible risk of explosion as a danger zone for proper control of the area. In the event where a fire, submersion, or collision involves high-voltage batteries, always place the vehicle in an open area at least 15 meters away from inflammables.

#### • Prevention of Reignition

Even after the high-voltage battery fire appears to have been extinguished, reignition or delayed ignition may occur. Use a thermal imaging camera to ensure the high-voltage battery is completely cooled before leaving the fire scene.

Advise subsequent emergency responders that there is a risk of battery fire reignition. Even after a fire suppression operation is completed, stand by on the scene for a while to monitor the situation.

#### • Caution Against Toxic Gas

A burning battery may release toxic gases, such as hydrogen fluoride, carbon monoxide, and carbon dioxide. Use self-contained breathing apparatus (SCBA) with full protective gear.

Since fire suppression operation can pose serious risks to the respiratory system, be sure to wear the SCBA at all times during the operation.

## 6. In case of fire



### ● Fire Extinguishing Kit

#### 1) Personal Protective Equipment (PPE)

- Chemical / high-temperature protective clothing : Wear gear to protect against chemicals and high temperature
- Electrical gloves and boots: Wear insulated gloves to avoid a high-voltage electrocution risk
- Safety helmets and goggles : Wear a helmet and goggles to protect against falling objects and from smoke
- Respiratory protective gear (e.g. SCBA, respirator)
  - : Wear respiratory protective gear to prevent inhalation of toxic gases and smoke

### **WARNING** Irritant Materials Caution !

- The high-voltage battery contains stimulants and sensitizers. Wear Personal Protective Equipment (PPE) and Self-Contained Breathing Apparatus (SCBA) to avoid contact with stimulants and sensitizers.
- Electrolyte solution is an eye-irritant substance. If the solution comes into contact with your eyes, rinse them [thoroughly](#).
- Electrolyte solution is a skin-irritant substance. If the solution comes into contact with your skin, wash off the affected area with soap and water.
- Electrolyte liquid or fumes coming into contact with water will create vapors in the air from oxidization. These vapors may [irritate](#) skin and eyes. In the event of contact with vapors, rinse the affected area with plenty of water and consult a doctor immediately.
- Electrolyte fumes can cause respiratory irritation and acute intoxication when inhaled. Inhale fresh air and rinse your mouth with water. Consult a doctor immediately.

#### 2) FCEV Fire Extinguishing Kit

- Fire blanket & portable water tank: To effectively suppress EV fires
- Thermal imaging camera: To monitor the temperature of batteries and the affected area
- Cutting tools (e.g. cutter, circuit breaker): To shut down the power and perform an extrication operation

## 6. In case of fire



### ● FCEV Fire Suppression

Control the fire scene and set a danger zone to limit the access of persons other than responders. FCEV battery fire has a risk of reignition. When extinguishing the fire, be sure to keep a safe distance from it. Use a large volume of water to prevent the spread of fire in the initial stage of response, and afterwards, use a portable water tank, etc. to put out the fire.

#### • **Spray a Large Volume of Water**

It is crucial to use a large volume of water to suppress FCEV battery fires since such type of fire is highly likely to be reignited. Use a high-pressure hose to spray a large volume of water directly onto the battery pack, or especially the area where flames are rising. In this way, you can inject water into the battery pack and effectively lower the heat.

Constantly spray water at least for tens of minutes up to a few hours until the battery is completely cooled down.

On average, 22,340 – 26,200 liters of water are required for FCEV fire suppression (The amount of water used for a water tank not included, source: National Fire Agency statistics).

#### • **Submerge the Vehicle in Water**

Suppressing the fire as part of the initial response, moving the vehicle to a safe place and submerging it in a container full of water or a portable water tank is the most effective way to extinguish fire. Leave the vehicle submerged for more than 72 hours so that the batteries are submerged and cooled. Depending on the type and size of the water tank, the amount of water required may vary from 5,000 to 14,300 liters (water tank size: 500x200x50 - 650x400x55).

### ● Things to Consider During Fire

#### **Suppression**

##### • **High-voltage Battery Pack Handling Caution**

Be careful not to physically access or impact the battery pack during fire suppression.

This may increase the risk of electrocution and explosion.

If there is an opening in the battery as a result of collision or damage, spraying directly onto the opening part will effectively extinguish a fire. However, never attempt to forcibly penetrate the battery to inject water into it.

##### • **Electrocution Risk**

Since spraying onto FCEV batteries may cause electrocution, fire responders should always wear protective gear and avoid direct contact with the battery.

##### • **Refrain from Use of Fire Extinguishing Agents Other Than Water**

Non-active fire extinguishing agents, such as CO<sub>2</sub>, powder extinguishing agent, or foam extinguishing agent, are not effective in suppressing EV fires, and may worsen the fire. The most effective way to put out such fire is to use water.

## 6. In case of fire



### ● Prevention of Reignition

Unlike other vehicle fires, FCEV fires have a higher risk of reignition even after they have been put out due to the characteristics of their high-voltage batteries. Therefore, follow-up measures are required after suppressing such fire to confirm whether the fire has been completely put out and to prevent reignition.

#### • Use a Thermal Imaging Camera

A thermal imaging camera can measure the temperature of the battery cell and the surrounding area without physically touching the target area. It is also useful for detecting heat of any internal area where it is difficult to see with the naked eye. Measure the temperature of the entire vehicle and the high-voltage battery pack repeatedly by using the thermal imaging camera to see if the temperature remains stable. If the fire is completely suppressed, the temperature should be even and stable across all areas measured.

#### • Standard for Judging Complete Fire Suppression

Submerge a battery in a water tank for at least 72 hours and use a thermal imaging camera to check the battery temperature. If the temperature remains stable for 1 hour, there is no likelihood of reignition.

In the event where using a thermal imaging camera is difficult, check whether no reignition occurs for a certain amount of time (at least 24 hours) after submersion.

### ● Post-Fire Suppression Operation

#### • Check Battery Safety

After putting out a fire, firefighters should confirm whether the fire has been successfully suppressed, and thoroughly confirm the stability of the battery pack and battery safety. If necessary, the vehicle should be inspected by a manufacturer or a professional technician.

#### • Move the Vehicle to a Safe Place

Move the vehicle, whose fire has been successfully extinguished, to a safe place to prevent any further risk.



## 6. In case of fire



### ● Response Measures When Exposed to Chemicals

In the event of inhaling electrolyte fumes or skin contact with electrolyte solutions, it is recommended that you consult a doctor even when the symptoms look mild.

#### 1) Skin Contact with Electrolyte Solutions

##### • Immediate Rinsing

In the event of skin contact with electrolyte solutions, immediately rinse with clean water for at least 15 minutes. Prompt rinsing is crucial since chemicals may penetrate the skin and cause a burn.

##### • First Aid and Getting Medical Help

If you have serious skin irritation or a burn on the skin, administer first aid and immediately get medical help.

#### 2) When Getting Electrolyte Solution in Your Eyes

##### • Immediate Rinsing

If you get electrolyte solution in your eyes, rinse your eyes with clean water for at least 15 minutes. When rinsing, open your eyes to the fullest extent and remove any contact lenses, etc.

##### • First Aid and Getting Medical Help

If you have a severe eye burn or irritation due to chemicals, immediately get medical help.

#### 3) Response Measures for Smoke Inhalation

##### • Inhaling Fresh Air

Inhalation of electrolyte fumes may cause respiratory irritation and acute intoxication. In case of inhaling electrolyte fumes, immediately leave the fire scene, breathe fresh air and rinse your mouth with water and consult a doctor.

##### • When Having Respiratory Symptoms

If you have respiratory symptoms, such as a cough, shortness of breath, or chest pain, immediately seek medical attention. If oxygen deficiency or serious toxic gas inhalation is suspected, you may need an oxygen mask under the instructions of the medical personnel.

## 7. In case of submersion

### Submerged Vehicle

Some emergency responses can involve a submerged vehicle. High voltage does not run through the submerged All-new NEXO body unless there is severe damage to the vehicle. In such case, it is safe to touch the vehicle body or framework – whether it is under water or on land. In the event where the vehicle is fully or partially submerged, retrieve the vehicle from water before attempting to disable the vehicle. Drain water from the vehicle. Use one of the methods described earlier to disable the vehicle.

#### WARNING



- If high-voltage components are exposed due to severe damage to the vehicle, responders should handle the high-voltage system in accordance with the precautions and wear appropriate personal protective equipment in order to prevent a safety accident.
- Do not remove the safety plug when the vehicle is under water.
- Failure to follow these instructions can lead to serious injury or death from electrocution.

## 8. Towing / transportation / storage

### Towing

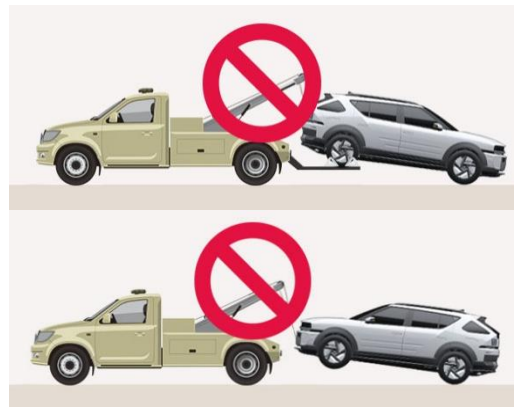
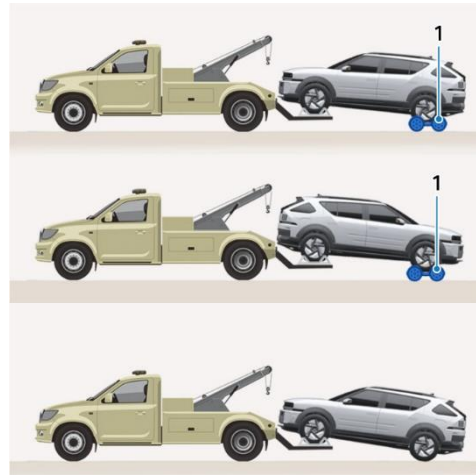
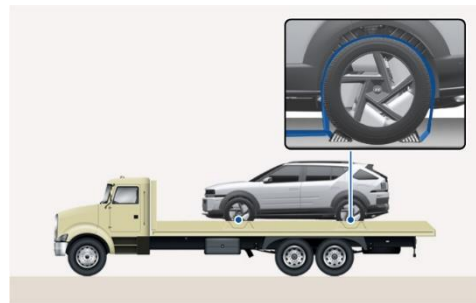
Towing the All-new NEXO is not different from towing other vehicles with a gasoline engine. If emergency towing is necessary, we recommend having it done by an authorized HYUNDAI dealer or a professional tow-truck service. Proper lifting and towing procedures are necessary to prevent damage to the vehicle. The best way to tow this vehicle is to lift the entire vehicle using a tow truck. However, when towing the vehicle using two wheels, use a flatbed or wheel dollies and lift the front wheels as the vehicle is equipped with a front wheel drive motor.

Before towing the vehicle, release the parking brake and turn the vehicle off. Then, press the 「P RELEASE」 button and put the shift lever in N (neutral). Detailed guidelines for vehicle towing are as follows.

#### WARNING



- Do not tow with sling-type equipment or with its front wheels touching the ground. It is not a proper way to tow this vehicle.
- Always use a wheel lift or flatbed equipment to prevent any damage to the vehicle.
- Not using the proper towing method can cause damage to the vehicle or additional accidents.
- When towing the vehicle equipped with side airbags, make sure to turn off the vehicle before towing. If the vehicle inclines with POWER ON, the system may detect it as a rollover, leading to deployment of side airbags.

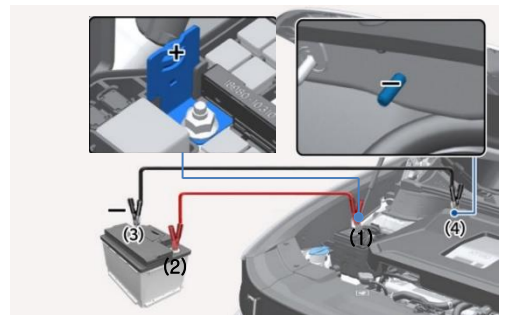
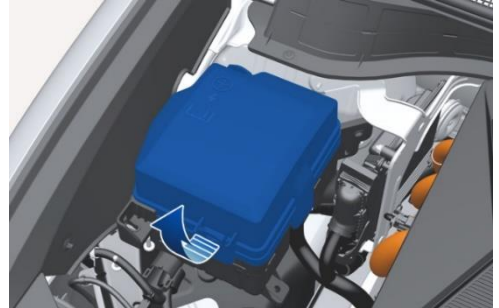


## 9. Important additional information

### Jump Starting

1. Make sure the booster battery is 12 V.
2. If the booster battery is in another vehicle, do not allow the vehicles to touch.
3. Turn off all unnecessary electrical devices.
4. First, open the fuse box located at the left side of the stack room. Then, connect one red clamp of the jumper cable to the positive (+) terminal ① of the dead battery and the other red clamp to the positive (+) terminal ② of the working battery.

Connect one black clamp to the negative (-) terminal ③ of the working battery and the other clamp to the negative (-) terminal ④ of the dead battery or to an unpainted metal part away from the positive (+) terminal of the dead vehicle (e.g. bolts at the lower part of a cowl or vehicle body).



#### WARNING



- Do not connect the cables to or near any part that moves when starting the vehicle.
- Do not allow the jumper cables to contact anything other than the correct battery terminals or the correct ground.
- Do not lean over the battery when connecting jumper cables.

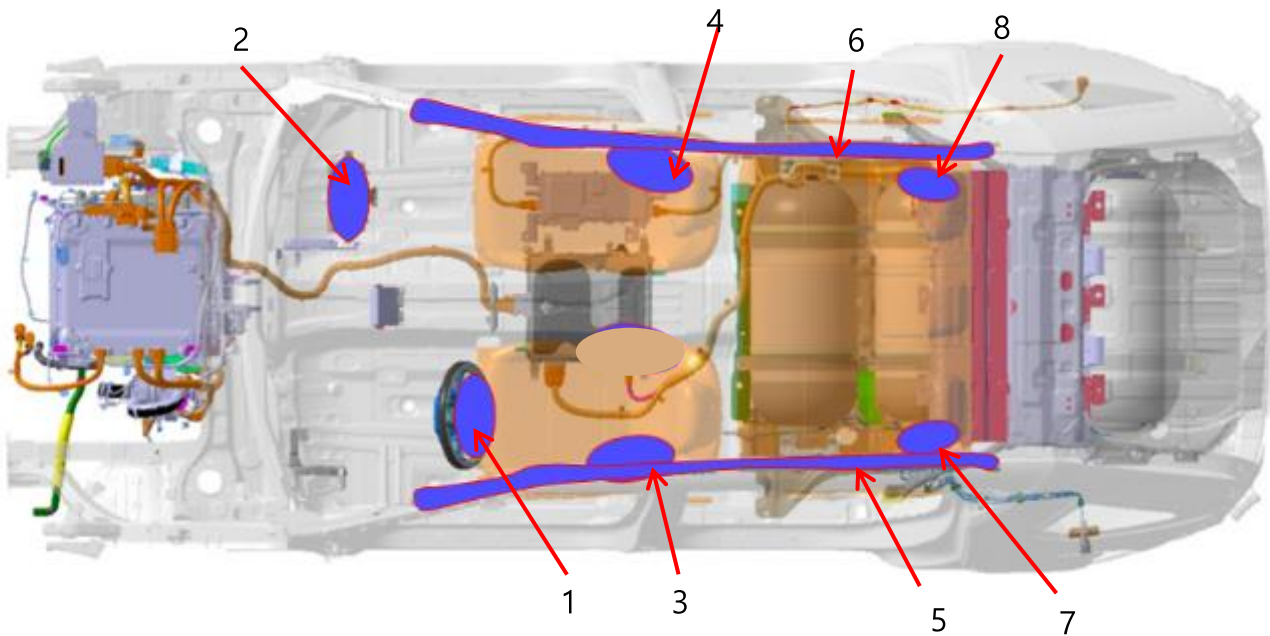
5. When connecting the jumper cables, start the working car first and let it run for a few minutes.
6. Press the 「12V BATT RESET」 button.
7. Turn on the discharged vehicle after 15 seconds. Remove the jumper cables after starting the vehicle.
8. Disconnect the (-) terminal cable first, and then the positive (+) terminal cable.
9. With the 「」 (READY) indicator on, stop or drive the vehicle for 30 minutes.
10. If the cause of battery discharge is not clear, we recommend that the system be checked by an authorized HYUNDAI dealer.



# 9. Important additional information

## Airbag System (SRS: Supplemental

**Restraint)** EXO is equipped with eight airbags as shown in the illustration below. Before starting any emergency procedure, make sure that the vehicle ignition is turned off and disconnect the negative cable from the 12 V auxiliary battery to prevent accidental deployment of airbags.



\* The actual airbags and seats in the vehicle may differ from the illustration.

| Callout | Name                          | Location                       |
|---------|-------------------------------|--------------------------------|
| 1       | Driver's side front airbag    | Driver's seat                  |
| 2       | Passenger's side front airbag | Passenger's seat               |
| 3,4     | Front side airbags            | Driver's seat/passenger's seat |
| 5,6     | Curtain airbags               | Driver's seat/passenger's seat |
| 7,8     | Rear side airbags             | Left/right of the rear seats   |

### WARNING



- Do not cut airbag components.
- Airbag components can remain powered and active for a maximum of 3 minutes even after the 12 V electrical system shuts down or is disconnected. Discharge the negative battery cable and wait 3 minutes or longer before starting any operation.

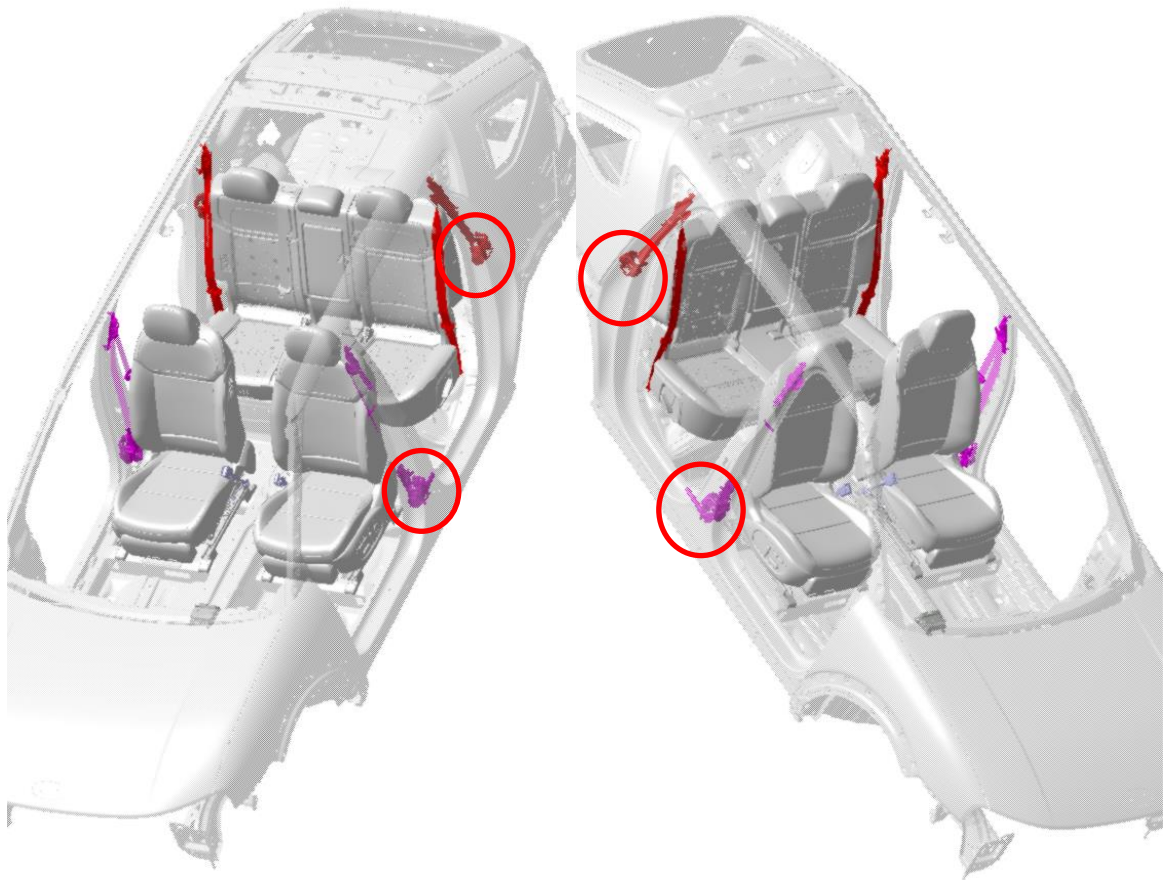
※ Failure to follow any of these instructions may result in serious injury or death from accidental airbag deployment.



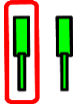
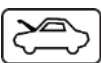
## 9. Important additional information

### Seat Belt Pretensioners

In the All-new NEXO, the driver's and front passenger's seat belts are equipped with pretensioners. When the seat belt pretensioners are activated during a collision, a loud noise may be heard and fine dust, which may seem like smoke, may be visible in the passenger compartment. These are normal operating conditions and are not hazardous. The seat belt pretensioners may become hot during activation and may need several minutes to cool down after they have been activated. The total of 4 pretensioners are installed in the All-new NEXO – with one seat belt pretensioner (BPT) and one anchor pretensioner (APT) at each B-pillar of the driver's and front passenger's seats.



## 10. Explanation of pictograms used

|                                                                                    |                                |                                                                                     |                                                                       |                                                                                     |                            |                                                                                       |                                                          |
|------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------|
|    | Airbag                         |    | Stored gas inflator                                                   |    | Seatbelt pretensioner      |    | SRS control unit                                         |
|    | Battery, low voltage           |    | Gas strut / Preloaded spring                                          |    | High strength zone         |    | Cable cut                                                |
|    | Battery pack, high-voltage     |    | High voltage power cable                                              |    | High voltage component     |    | Use thermal infrared camera                              |
|    | Disconnect high voltage device |    | Gas tank with gas type indication                                     |    | Fuel cell component        |    | Use water to extinguish the fire                         |
|    | General warning sign           |    | Warning, Electricity                                                  |    | Environmental hazard       |    | Explosive                                                |
|   | Acute toxicity                 |   | Corrosives                                                            |   | Bonnet                     |   | Boot                                                     |
|  | Flammable                      |  | Hazardous to the human health                                         |  | Fuel Cell Electric Vehicle |  | Attention; hydrogen burns with an almost invisible flame |
|  | Disconnect high voltage device |  | Automatic hydrogen overpressure safety valve with gas type indication |                                                                                     |                            |                                                                                       |                                                          |